

Web Technologies

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Explore the package `jakarta.servlet` and `jakarta.servlet.html` packages for their classes, methods, interfaces with their prototype.

The **`jakarta.servlet`** package forms the core of the Jakarta Servlet API, providing the fundamental interfaces and classes for developing servlets. The **`jakarta.servlet.http`** package, a sub-package, extends this functionality specifically for HTTP-based servlets. The user mentions `jakarta.servlet.html` but this package does not exist as a standard part of the Jakarta Servlet specification. HTML generation is typically handled within servlets using output streams or through technologies like Jakarta Server Pages (JSP), which are built on top of servlets.

Key Interfaces:

Servlet: The central interface that all servlets must implement. It defines lifecycle methods:

`void init(ServletConfig config):` Initializes the servlet.

`void service(ServletRequest req, ServletResponse res):` Handles client requests.

`void destroy():` Cleans up resources.

`ServletConfig getServletConfig():` Returns the `ServletConfig` object.

`String getServletInfo():` Returns information about the servlet.

ServletConfig: Provides initialization parameters and a reference to the ServletContext.

String getInitParameter(String name): Retrieves an initialization parameter.

Enumeration<String> getInitParameterNames(): Returns all initialization parameter names.

ServletContext getServletContext(): Returns the ServletContext.

String getServletName(): Returns the servlet's name.

ServletContext: Defines methods for a servlet to communicate with its container.

Object getAttribute(String name): Retrieves an attribute.

void setAttribute(String name, Object object): Sets an attribute.

void removeAttribute(String name): Removes an attribute.

RequestDispatcher getRequestDispatcher(String path): Returns a RequestDispatcher.

String getMimeType(String file): Returns the MIME type of a file.

void log(String message): Writes a log message.

ServletRequest: Represents a client request.

String getParameter(String name): Retrieves a request parameter.

InputStream getInputStream(): Returns an input stream for binary data.

ServletResponse: Represents the servlet's response to the client.

PrintWriter getWriter(): Returns a PrintWriter for character data.

ServletOutputStream getOutputStream(): Returns an output stream for binary data.

void setContentType(String type): Sets the content type of the response.

Key Classes:

GenericServlet: An abstract class implementing Servlet and ServletConfig, providing default implementations for most methods, making it easier to write servlets.

jakarta.servlet.http Package

Key Interfaces:

HttpServletRequest: Extends ServletRequest with HTTP-specific methods.

String getMethod(): Returns the HTTP method (GET, POST, etc.).

HttpSession getSession(): Returns the current HttpSession.

Cookie[] getCookies(): Returns an array of Cookie objects.

HttpServletResponse: Extends ServletResponse with HTTP-specific methods.

void addCookie(Cookie cookie): Adds a Cookie to the response.

void sendRedirect(String location): Sends a redirect response.

void setStatus(int sc): Sets the HTTP status code.

HttpSession: Represents a session between a client and a server.

Object getAttribute(String name): Retrieves an attribute from the session.

void setAttribute(String name, Object value): Binds an object to the session.

Key Classes:

HttpServlet: An abstract class extending `GenericServlet`, specifically designed for HTTP servlets. It provides `doGet()`, `doPost()`, `doPut()`, `doDelete()`, etc., methods to handle specific HTTP methods.

Cookie: Represents an HTTP cookie.

jakarta.servlet.html (Non-existent)

As mentioned, `jakarta.servlet.html` is not a standard package. HTML generation is typically handled within the `jakarta.servlet` or `jakarta.servlet.http` context by:

Writing HTML directly to the PrintWriter or ServletOutputStream: obtained from `ServletResponse` or `HttpServletResponse`

Java

```
response.setContentType("text/html");

PrintWriter out = response.getWriter();

out.println("<html><body><h1>Hello
World!</h1></body></html>");
```

Using Jakarta Server Pages (JSP), which allows embedding Java code within HTML-like pages, and are compiled into servlets by the container.

```
package jakarta.servlet
```

The `jakarta.servlet` package contains a number of classes and interfaces that describe and define the contracts between a servlet class and the runtime environment provided for an instance of such a class by a conforming servlet container.

Related Packages

Package

Description

<code>jakarta.servlet.annotation</code>

<code>jakarta.servlet.descriptor</code>

<code>jakarta.servlet.http</code>

The `jakarta.servlet.http` package contains a number of classes and interfaces that describe and define the contracts between a servlet class running under the HTTP protocol and the runtime environment provided for an instance of such a class by a conforming servlet container.

Class

Description

AsyncContext

Provides the context for asynchronous request handling

AsyncEvent

Used to pass data to the methods of [AsyncListener](#).

AsyncListener

Listener for events associated with an [AsyncContext](#).

DispatcherType

Enumeration of dispatcher types.

Filter

A filter is an object that performs filtering tasks on either the request to a resource (a servlet or static content), or on the response from a resource, or both.

FilterChain

A FilterChain is an object provided by the servlet container to the developer giving a view into the invocation chain of a filtered request for a resource.

FilterConfig

A filter configuration object used by a servlet container to pass information to a filter during initialization.

FilterRegistration

Interface through which a Filter may be further configured.

FilterRegistration.Dynamic

Interface through which a Filter registered via one of the addFilter methods on ServletContext may be further configured.

GenericFilter

Provides a base class that implements the `Filter` and `FilterConfig` interfaces to reduce boilerplate when writing new filters.

GenericServlet

Defines a generic, protocol-independent servlet.

HttpConstraintElement

Equivalent of `HttpConstraint` for programmatic configuration of security constraints.

HttpMethodConstraintElement

Programmatic equivalent of a security constraint defined for a single HTTP method.

MultipartConfigElement

The programmatic equivalent of `MultipartConfig` used to configure multipart handling for a Servlet when registering a Servlet via code.

ReadListener

Receives notification of read events when using non-blocking IO.

Registration

Common interface for the registration of Filters and Servlets.

Registration.Dynamic

Interface through which a Servlet or Filter registered via one of the `addServlet` or `addFilter` methods, respectively, on `ServletContext` may be further configured.

RequestDispatcher

Defines an object that receives requests from the client and sends them to any resource (such as a servlet, HTML file, or JSP file) on the server.

Servlet

Defines methods that all servlets must implement.

ServletConfig

A servlet configuration object used by a servlet container to pass information to a servlet during initialization.

ServletConnection

Provides information about the connection made to the Servlet container.

ServletContainerInitializer

ServletContainerInitializers (SCIs) are registered via an entry in the file META-INF/services/jakarta.servlet.ServletContainerInitializer that must be included in the JAR file that contains the SCI implementation.

ServletContext

Defines a set of methods that a servlet uses to communicate with its servlet container, for example, to get the MIME type of a file, dispatch requests, or write to a log file.

ServletContextAttributeEvent

This is the event class for notifications about changes to the attributes of the servlet context of a web application.

ServletContextAttributeListener

Implementations of this interface receive notifications of changes to the attribute list on the servlet context of a web application.

ServletContextEvent

This is the event class for notifications about changes to the servlet context of a web application.

ServletContextListener

Implementations of this interface receive notifications about changes to the servlet context of the web application they are part of.

ServletException

Defines a general exception a servlet can throw when it encounters difficulty.

ServletInputStream

Provides an input stream for reading binary data from a client request, including an efficient readLine method for reading data one line at a time.

ServletOutputStream

Provides an output stream for sending binary data to the client.

ServletRegistration

Interface through which a Servlet may be further configured.

ServletRegistration.Dynamic

Interface through which a Servlet registered via one of the addServlet methods on ServletContext may be further configured.

ServletRequest

Defines an object to provide client request information to a servlet.

ServletRequestAttributeEvent

This is the event class for notifications of changes to the attributes of the servlet request in an application.

ServletRequestAttributeListener

A ServletRequestAttributeListener can be implemented by the developer interested in being notified of request attribute changes.

ServletRequestEvent

Events of this kind indicate lifecycle events for a ServletRequest.

ServletRequestListener

A ServletRequestListener can be implemented by the developer interested in being notified of requests coming in and out of scope in a web component.

ServletRequestWrapper

Provides a convenient implementation of the ServletRequest interface that can be subclassed by developers wishing to adapt the request to a Servlet.

ServletResponse

Defines an object to assist a servlet in sending a response to the client.

ServletResponseWrapper

Provides a convenient implementation of the ServletResponse interface that can be subclassed by developers wishing to adapt the response from a Servlet.

ServletSecurityElement

The programmatic equivalent of [ServletSecurity](#) used to configure security constraints for a Servlet.

SessionCookieConfig

Configures the session cookies used by the web application associated with the ServletContext from which this SessionCookieConfig was obtained.

SessionTrackingMode

Defines the permitted options for configuring the session tracking mode.

UnavailableException

Defines an exception that a servlet or filter throws to indicate that it is permanently or temporarily unavailable.

Package jakarta.servlet.http

The jakarta.servlet.http package contains a number of classes and interfaces that describe and define the contracts between a servlet class running under the HTTP protocol and the runtime environment provided for an instance of such a class by a conforming servlet container.

Interface Summary

Interface	Description
HttpServletMapping	Allows runtime discovery of the manner in which the HttpServlet for the current HttpServletRequest was invoked.
HttpServletRequest	Extends the ServletRequest interface to provide request information for HTTP servlets.
HttpServletResponse	Extends the ServletResponse interface to provide HTTP-specific functionality in sending a response.
HttpSession	Provides a way to identify a user across more than one page request or visit to a Web site and to store information about that user.

HttpSessionActivationListener	Objects that are bound to a session may listen to container events notifying them that sessions will be passivated and that session will be activated.
HttpSessionAttributeListener	Interface for receiving notification events about HttpSession attribute changes.
HttpSessionBindingListener	Causes an object to be notified when it is bound to or unbound from a session.
HttpSessionContext	Deprecated <i>As of Java(tm) Servlet API 2.1 for security reasons, with no replacement.</i>
HttpSessionIdListener	Interface for receiving notification events about HttpSession id changes.
HttpSessionListener	Interface for receiving notification events about HttpSession lifecycle changes.
HttpUpgradeHandler	This interface encapsulates the upgrade protocol processing.
Part	This class represents a part or form item that was received within a multipart/form-data POST request.
PushBuilder	Build a request to be pushed.
WebConnection	This interface encapsulates the connection for an upgrade request.

Class Summary

Class	Description
Cookie	Creates a cookie, a small amount of information sent by a servlet to a Web browser, saved by the browser, and later sent back to the server.
HttpFilter	Provides an abstract class to be subclassed to create an HTTP filter suitable for a Web site.

HttpServlet	Provides an abstract class to be subclassed to create an HTTP servlet suitable for a Web site.
HttpServletRequestWrapper	Provides a convenient implementation of the HttpServletRequest interface that can be subclassed by developers wishing to adapt the request to a Servlet.
HttpServletResponseWrapper	Provides a convenient implementation of the HttpServletResponse interface that can be subclassed by developers wishing to adapt the response from a Servlet.
HttpSessionBindingEvent	Events of this type are either sent to an object that implements HttpSessionBindingListener when it is bound or unbound from a session, or to a HttpSessionAttributeListener that has been configured in the deployment descriptor when any attribute is bound, unbound or replaced in a session.
HttpSessionEvent	This is the class representing event notifications for changes to sessions within a web application.
HttpUtils	Deprecated <i>As of Java(tm) Servlet API 2.3.</i>

